

Few-shot foundation-model mixture of experts for cereal mapping in data-scarce regions

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Abstract. We present a data-efficient pipeline for semantic segmentation of cereal crops from Sentinel-2 imagery in Algeria. Leveraging two pretrained Vision Foundation Models (Prithvi-EO-2.0 and Satlas), we fine-tuned both under a few-shot regime (124 spatially disjoint image-mask pairs collected in Sétif) and combined them through a lightweight Mixture-of-Experts (MoE) gating network. On a held-out validation split, we report mean Cereal-class Intersection-over-Union (IoU) = 95.6 ± 0.9 % and Overall Accuracy = 97.8 ± 0.6 % (mean $\pm$ SD over 3 independent runs). Detailed ablations show that the MoE improves IoU by +1.5–2.0 % relative to the best single expert, and that RandAugment and band normalization are critical for training stability. We release code, model checkpoints, and a dataset sample to enable reproducibility. This work demonstrates that foundation-model fusion via MoE yields robust cereal mapping under severe label scarcity, with potential for transfer to other crops and regions.

Keywords: foundation models, few-shot learning, mixture of experts, crop mapping, Sentinel-2, semantic segmentation.

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1 Introduction

Agriculture represents a vital component of Algeria’s economy, contributing approximately 12% to the national GDP and employing nearly 25% of the workforce. However, the sector faces persistent challenges such as water scarcity, pest damage, and inefficient resource management, which collectively account for an estimated 20-30% loss of potential crop yields annually.^{1,2} These constraints threaten national food security and underscore the need for innovative, technology-driven solutions.³ Integrating remote sensing (RS) with artificial intelligence (AI) provides a transformative pathway for precision agriculture,⁴ enabling data-driven monitoring of crop health

and environmental conditions in real time.⁵ Nevertheless, the effectiveness of such AI-based monitoring remains limited by the scarcity of locally labeled datasets suitable for training robust deep learning models.

A fundamental impediment to deploying advanced AI in Algerian agriculture is the critical scarcity of labeled national datasets.⁶⁻⁸ Given the limited availability of task-specific annotated data, developing deep learning models from scratch becomes impractical.^{9,10} This data gap is further compounded by the high computational cost of training, with some models requiring over 100 GPU hours. These constraints strongly advocate for the use of data-efficient techniques such as transfer learning, which can leverage knowledge from large, pre-trained Vision Foundation Models (VFMs).^{11,12}

To address these limitations, we collected a local dataset of 124 Sentinel-2 image-mask pairs representing cereal and non-cereal parcels in the Sétif region, using the Mergin Maps platform for in-field polygon delineation between March and May 2024 (see Sec. 3.2). Each polygon was labeled as either 1 (cereal) or 0 (non-cereal), producing a binary segmentation dataset suitable for few-shot learning. This study directly explores the underutilized potential of fine-tuning VFMs through Few-Shot Learning (FSL)—a training approach that adapts models using only a small number of labeled examples—for robust semantic segmentation of crop types under severe data scarcity.^{13,14} By developing accessible and computationally efficient monitoring solutions, our work aims to mitigate inefficiencies in water and fertilizer management and support sustainable agricultural planning in resource-constrained environments.

The main contributions of this study are summarized as follows:

- We collected and curated a local Sentinel-2 dataset covering cereal crop parcels in the Sétif region of Algeria, providing 124 spatially disjoint image-mask pairs annotated for cereal versus non-cereal areas under severe data scarcity conditions.
- We fine-tuned two Vision Foundation Models, Prithvi-EO-2.0 and Satlas, using a few-shot learning strategy designed for limited samples while maintaining generalization across unseen parcels.
- We developed a lightweight Mixture-of-Experts (MoE) gating network that adaptively fuses predictions from both foundation models, yielding improved semantic segmentation performance and enhanced robustness.

- We achieved a mean Intersection-over-Union (IoU) of 95.6 ± 0.9 % and an Overall Accuracy of 97.8 ± 0.6 % (mean $\pm$ SD over 3 independent runs), surpassing single-model baselines by up to 2 %. All code, trained weights, and a sample of the dataset are released to ensure reproducibility and future research reuse.

2 Related Work

2.1 Foundation models for Earth observation

This section describes the normal structure of a manuscript and how each part should be handled. Recent years have seen the rapid emergence of Vision Foundation Models (VFMs) tailored to Earth observation (EO), pretrained on very large collections of satellite and aerial imagery and then adapted to downstream tasks via transfer learning or fine-tuning. Large-scale pretraining efforts such as SatlasPretrain have shown that pretraining on diverse, geospatially distributed imagery substantially improves downstream performance on segmentation and mapping tasks.¹⁵ New multi- temporal foundation models (e.g., Prithvi-EO-2.0) further integrate temporal and location embed- dings to boost generalization across sensors and phenological stages, making them particularly suitable for crop-mapping tasks.¹⁶ Satellite-specific contrastive and vision-language approaches (e.g., SatCLIP) provide location-aware and open-vocabulary capabilities that enable stronger few- shot transfer in EO domains.¹⁷ Transformer-based backbones and hybrid vision-time architectures (GeoViT and GeoFormer) have also become popular choices for encoding the spatial and temporal structure of RS imagery.^{18,19}

2.2 Few-shot learning in remote sensing

Few-shot learning (FSL) and generalized few-shot semantic segmentation (GFSS) have received increasing attention in the RS community due to the persistent scarcity of dense annotations. Several recent benchmarks and methodological advances propose task-augmentation, prototype- and adapter-based strategies that leverage foundation models or learnable prompts to bridge domain gaps with very few labels.^{20,21} Domain-specific techniques in RS—including label propagation, feature disentanglement, and multi-scale prototype aggregation—have delivered strong results on aerial and spaceborne few-shot segmentation benchmarks (iSAID-5i, OpenEarthMap variants) and motivate the use of few-shot fine-tuning for cereal mapping under severe label scarcity.¹³

2.3 Ensembles, Mixture-of-Experts, and diffusion-based approaches

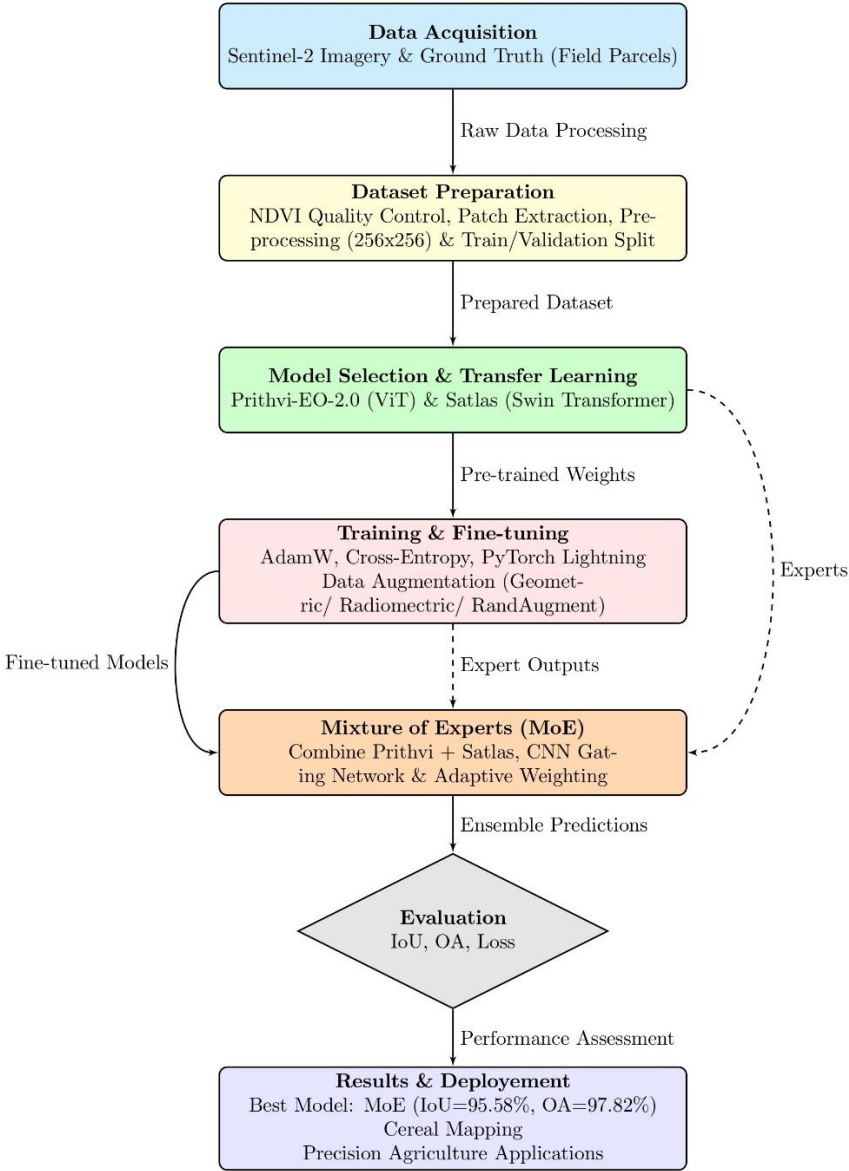
Ensembling and Mixture-of-Experts (MoE) are natural approaches to combine complementary models and improve robustness for segmentation tasks. Classical ensembling (e.g., U-Net ensembles with diverse backbones) and lightweight gated fusion schemes have been shown to increase segmentation accuracy in RS contexts.^{22,23} At the model-architecture level, MoE ideas have matured from sparsely gated large networks to practical, task-adaptive gating layers that can combine pretrained experts with small additional capacity.²⁴ Separately, diffusion-based generative and restoration models have quickly become important tools in RS for image synthesis, cloud removal, and spatiotemporal fusion; they also offer promising avenues for data augmentation and out-of-distribution detection that can complement few-shot pipelines.^{25–27}

Table 1 Recent works (2023–2025) related to foundation models, few-shot learning, and ensemble/MoE and diffusion methods in remote sensing.

Paper	Dataset(s)	Pretraining/ Modality	Task / Metric	Key novelty
SatlasPretrain ¹⁵	Satlas (302M labels)	Large-scale RS pretraining	Multi-task transfer	Multi-task RS pretraining; strong transfer gains.
Prithvi-EO-2.0 ¹⁶	Harmonized Landsat + Sentinel	Multi-temporal pretraining	Segmentation / IoU	Temporal + location embeddings; multi-resolution generalization.
SatCLIP ¹⁷	Global imagery	CLIP-style embeddings	Zero/few-shot transfer	Location-aware CLIP embeddings for RS.
GeoViT ¹⁸	Sentinel/ multi-sensor	Transformer backbone	Segmentation / Accuracy	Vision transformer architecture for RS.
GeoFormer ¹⁹	Sentinel	Transformer backbone	Classification / IoU	Temporal transformer for RS.
GFSS ²⁰	OpenEarthMap	—	Few-shot segmentation	Benchmark for generalized few-shot segmentation.
FM-Assisted FSL ²¹	OpenEarthMap	FM-assisted	GFSS (mIoU)	Foundation-model-assisted few-shot segmentation.
U-Net Ensemble ²²	Various RS datasets	Ensembles	Segmentation	Ensemble gains via backbone diversity.

RSDiff ²⁵	Various RS imagery	Diffusion-based generation	Image synthesis / SR	Two-stage diffusion for high-res RS image synthesis.
CRS-Diff ²⁶	RS benchmarks	Diffusion tailored to RS	Conditional generation	Controllable diffusion for RS image generation.
STFDiff ²⁷	Spatiotemporal RS	Diffusion spatiotemporal	Fusion / reconstruction	Diffusion-based spatiotemporal fusion for RS.

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Fig. 1 General workflow of the proposed methodology.

3 Methodology

Figure 1 provides an overview of the proposed methodology. The workflow begins with Sentinel-2 imagery acquisition during the field campaign conducted between March and May 2024, corresponding to the late tillering to early heading stages of cereal crops in the Sétif region. Field data were collected simultaneously using the Mergin Maps mobile app to delineate cereal and non-cereal parcels. Sentinel-2 Level-2A images from April and May 2024 were selected for their minimal cloud coverage and optimal phenological representation. The workflow proceeds with NDVI-based quality control and preprocessing. The processed dataset is then used to fine-tune two vision foundation models (Prithvi and Satlas), and finally, a Mixture-of-Experts (MoE) ensemble combines both models to generate accurate cereal mapping results. The detailed steps are described in the following subsections.

3.1 Study Area and Data Source

The primary study area is centered around Sétif (Fig. 2), a key agricultural zone in the Hautes Plaines region of northeastern Algeria, recognized for its significant cereal production. The study area spans latitudes 36.14°N–36.29°N and longitudes 5.27°E–5.48°E, covering part of the Sétif high plain at an average elevation of approximately 1,080 m above sea level. The region is characterized by a semi-arid Mediterranean climate (Köppen classification: Csa) with annual precipitation between 350–450 mm. The local agricultural calendar includes cereal sowing from November to December and harvesting between late May and June. Sentinel-2 Level-2A images were selected from April to May 2024, coinciding with the late tillering to early heading stages of cereals, when canopy contrast is optimal for spectral discrimination. Field surveys were conducted between March and May 2024 using the Mergin Maps mobile application to collect 124 annotated polygon parcels distinguishing cereal and non-cereal fields.

The remote sensing data were sourced from the Sentinel-2 mission,²⁸ utilizing Level-2A atmospherically corrected surface reflectance products accessed via Google Earth Engine (GEE). Six spectral bands were selected for their sensitivity to vegetation: Blue (B2), Green (B3), Red (B4), Near-Infrared (NIR, B8), and Short-Wave Infrared (SWIR1–B11, SWIR2–B12). These bands provide complementary information for crop monitoring, as they capture visible reflectance linked to plant chlorophyll, near-infrared sensitivity to vegetation structure and biomass, and short-wave infrared responses to leaf water content and soil moisture.²⁹ Red-edge bands (B5, B6, B7,

B8A) were excluded to optimize computational efficiency, as prior studies showing that these bands provide only marginal improvements for crop-mapping tasks when NIR and SWIR channels are already included.^{30–32} All Sentinel-2 scenes were atmospherically corrected and resampled to 10 m spatial resolution using bilinear interpolation.

Table 2 Study area metadata.

Parameter	Value
Geographic extent	36.14°N–36.29°N, 5.27°E–5.48°E
Elevation	≈1080 m above sea level
Climate	Semi-arid Mediterranean (Csa)
Mean annual rainfall	350–450 mm
Main crop season	Nov–Jun (sowing: Nov–Dec, harvest: May–Jun)
Imagery period	Apr–May 2024
Field campaign	Mar–May 2024 (42 labeled parcels)

Figure 2 displays the mapped cereal parcels over the Sétif region, with geographic coordinates and scale reference.

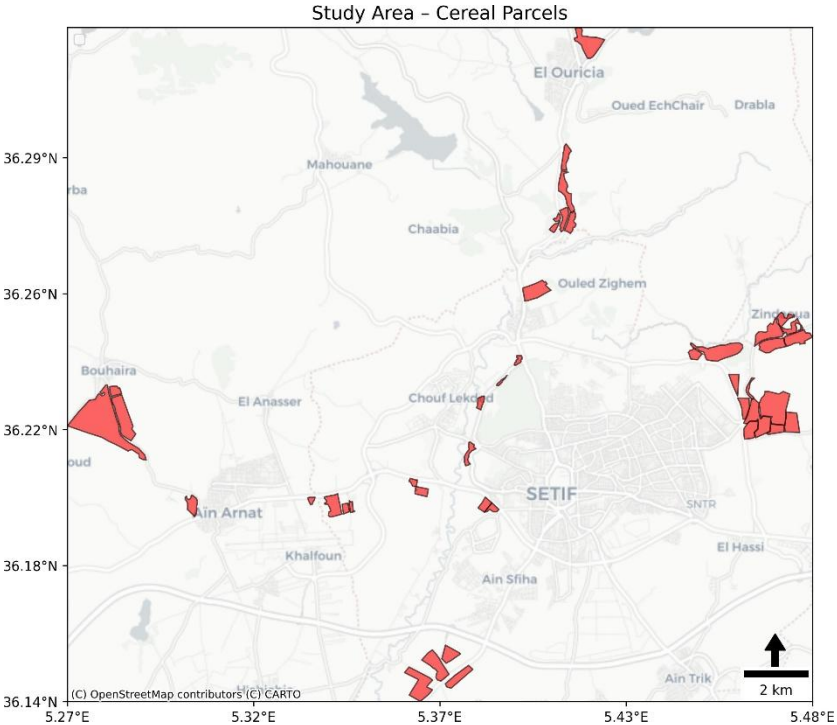


Fig. 2 The primary study area in Sétif, Algeria. The ground truth cereal parcel polygons collected for this research are highlighted in red.

3.2 Dataset Preparation and Preprocessing

Ground truth data were meticulously collected via three field campaigns conducted between March and May 2024 in Sétif using the Mergin Maps mobile application.³³ A team of two operators digitized cereal parcel polygons directly in the field using GPS-enabled smartphones with an average horizontal accuracy of ± 2 m.

Vegetation health and spectral dynamics were evaluated using time-series analysis of Sentinel-2 Level-2A surface reflectance products. Three vegetation indices were computed to ensure phenological and spectral consistency. The Normalized Difference Vegetation Index (NDVI) was calculated using the Near-Infrared (B8) and Red (B4) bands to assess vegetation vigor. The Normalized Difference Red Edge (NDRE) was employed to verify chlorophyll-content stability and identify early-senescing parcels, while the Enhanced Vegetation Index (EVI2) was used to ensure spectral consistency under varying illumination conditions. Polygons were retained only if $\text{NDVI} > 0.45$ and $\text{EVI2} > 0.35$ during the mid-season growth phase (April–May 2024). This multi-index filtering ensured that all parcels exhibited consistent vegetation vigor typical of active cereal stands.

The initial dataset contained 42 polygons. These underwent a rigorous quality assurance (QA) process based on geometric consistency, field notes, and the multi-index analysis described above. Six polygons exhibiting boundary ambiguity or anomalous phenological signatures were removed, yielding a final set of 36 validated polygons. To create the final dataset for training, these 36 polygons were sampled across multiple dates during the peak growing season, resulting in 124 distinct, spatially disjoint image-mask pairs. Each polygon contributed between 2–5 temporal samples representing different acquisition dates, resulting in a temporally rich yet spatially non-overlapping dataset.

Finally, images and masks were cropped and resized to 256×256 pixels. Data augmentation—including random rotations, flips, and brightness jittering—was applied to improve model robustness.

All pixel values were normalized per spectral band using the mean (μ) and standard deviation (σ) computed over the training set, according to Eq. 1:

$$x'_c = \frac{x_c - \mu_c}{\sigma_c}, \quad (1)$$

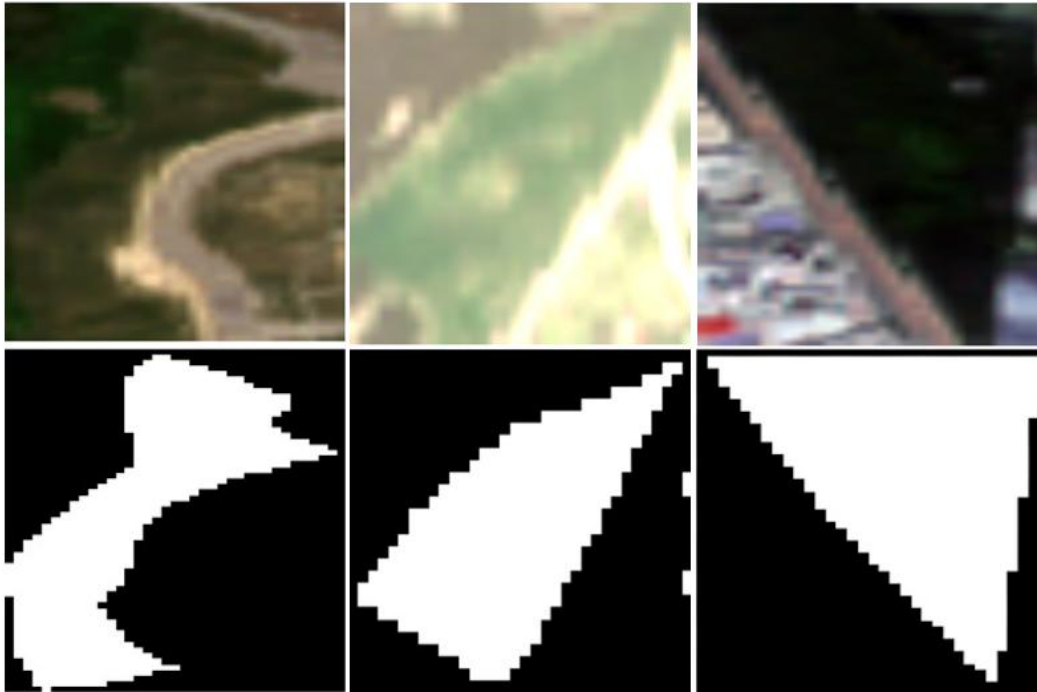
where x_c denotes the pixel intensity in channel c . Table 3 lists the per-band statistics used for normalization. All bands were resampled to a uniform 10 m resolution.

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Table 3 Per-band reflectance statistics used for normalization.

Band	Mean (μ)	Std (σ)
Blue (B2)	0.094	0.028
Green (B3)	0.107	0.031
Red (B4)	0.081	0.025
NIR (B8)	0.312	0.072
SWIR1 (B11)	0.198	0.057
SWIR2 (B12)	0.142	0.048

177 The dataset was divided into 80% for training and 20% for validation using a polygon-level split.
 178 Crucially, all temporal acquisitions derived from a specific ground truth polygon were assigned
 179 exclusively to the same split (either training or validation). This protocol ensures zero spatial or
 180 temporal leakage between sets, maintaining the strict independence required for valid few-shot
 181 evaluation. To evaluate model stability, experiments were repeated across three independent runs,
 182 and performance metrics (IoU, OA) were reported as mean $\pm$ standard deviation.



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184 **Fig. 3** Examples of a training samples, showing the input Sentinel-2 image patch (top) and its corresponding binary
 185 ground truth mask (bottom), where white represents the 'cereal' class.

3.3 Foundation Models and Fine-Tuning

This research adopted a transfer learning approach to overcome the limitations of a modest dataset size.³⁴ Two Vision Foundation Models (VFMs) were selected and fine-tuned for semantic segmentation:

3.3.1 Prithvi-EO-2.0

A ViT-based Earth Observation foundation model available in multiple sizes (notably 300M and 600M parameter variants). Prithvi-EO-2.0 is pretrained on multi-temporal harmonized Landsat+Sentinel-2 sequences (4.2M global time-series samples, ~ 30 m resolution) and incorporates explicit temporal and geolocation embeddings to model seasonal and geographic variability.¹⁶ The model replaces standard 2D ViT patch embeddings with 3D spatiotemporal patches and uses 3D positional encodings generated from separate 1D sin/cos functions for time, height, and width. During pretraining, metadata such as acquisition date and location are encoded and added to token embeddings through learned weights, with random metadata-dropping applied to increase robustness. These architectural elements make Prithvi especially suitable for tasks where temporal dynamics (crop phenology) are informative. Although Prithvi-EO-2.0 is pre-trained on temporal sequences, this study utilizes a single-temporal frame ($T = 1$) fine-tuning strategy. The model's 3D patch embeddings were adapted by treating the input as a sequence of length 1, focusing on spatial-spectral features extraction.

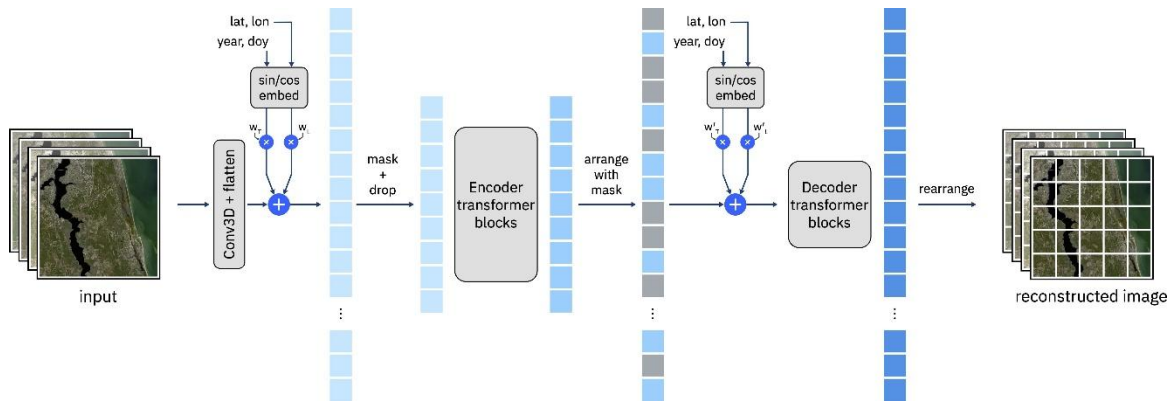


Fig. 4 Architecture of Prithvi-EO-2.0.¹⁶ The model uses ViT-based masked autoencoder pretraining with 3D patch embeddings to jointly encode spatial and temporal structure. Geolocation and temporal metadata are added through learned positional encodings.

3.3.2 Satlas (SatlasPretrain models)

The models follow a modular Swin-Transformer architecture composed of a backbone, a feature pyramid network (FPN), and a prediction head.¹⁵ The pretraining dataset combines Sentinel-2 and the US National Agriculture Imagery Program (NAIP) imagery paired with approximately 302M labels across 137 categories, generated from sources such as OpenStreetMap and WorldCover together with curated annotations. For multi-image (temporal) inputs, the backbone processes each image independently and temporal max pooling aggregates representations across time. This supervised, large-scale pretraining strategy enables Satlas to learn strong spatial priors, feature hierarchies, and semantic regularities that transfer effectively to dense prediction tasks such as segmentation. To align with the pre-trained Satlas Sentinel-2 weights, our 6-band input was mapped to the required 9-channel format via a specific transformation $T(x)$:

- Channels 0-2 (RGB): B4, B3, B2.
- Channels 3-5: Duplicates of B4, B3, B2 (to satisfy architectural requirements).
- Channel 6: B8 (NIR).
- Channels 7-8: B11, B12 (SWIR). Input values were normalized and clipping to [0, 1].

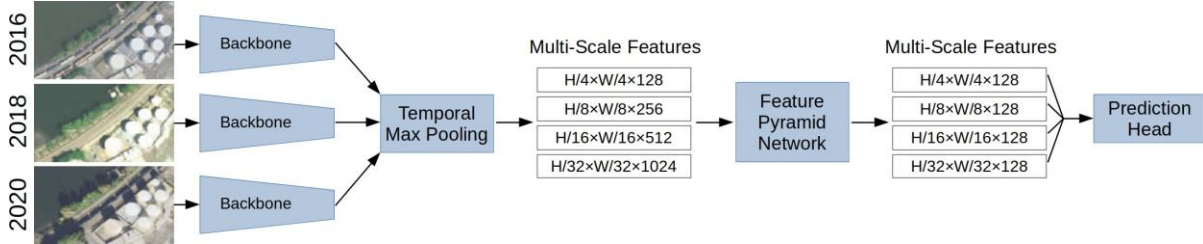


Fig. 5 Architecture of the SatlasPretrain model.¹⁵ The backbone extracts spatial features from each input image, an FPN builds multi-scale feature representations, and the prediction head produces dense outputs. Multi-image inputs are aggregated using temporal max pooling.

Both foundation models provide complementary strengths: Prithvi captures temporal cues through explicit multi-date modeling, while Satlas provides strong supervised semantic representations due to its label-rich pretraining. In this study, both models were initialized from publicly available pretrained weights and fine-tuned on the Sentinel-2 training set described earlier.

Formally, the segmentation network f_θ maps an input image patch $\mathbf{x}$ to per-pixel probabilities $\mathbf{p}$, with binary predictions $\hat{y}$ obtained as shown in Eq. 2:

$$\mathbf{p} = f_\theta(\mathbf{x}) \in [0,1]^{H \times W}, \quad \hat{y}_{ij} = \mathbf{1}\{p_{ij} \geq 0.5\}, \quad (2)$$

Training was conducted using the AdamW optimizer and the cross-entropy loss function (Eq. 3) within a PyTorch Lightning framework:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W [y_{ij} \log p_{ij} + (1 - y_{ij}) \log(1 - p_{ij})], \quad (3)$$

Where $H \times W$ is the image patch size, $y_{ij} \in \{0,1\}$ is the ground truth label at pixel (i, j) , and $p_{ij} \in [0,1]$ is the model’s predicted probability of that pixel belonging to the cereal class. The loss penalizes confident but wrong predictions (e.g., $y_{ij} = 0$ but $p_{ij} \approx 1$).

While other studies have found success with CNN-based architectures like ResNet³⁵ and VGG,³⁶ this work focuses on Transformer-based VFMs. All training hyperparameters used for fine-tuning the Prithvi and Satlas models are summarized in Table 4.

Table 4 Training hyperparameters used for fine-tuning the Prithvi and Satlas models.

Hyperparameter	Value
Learning rate	1e-4
Batch size	8
Number of epochs	50
Optimizer	AdamW
Weight decay	1e-2
Loss function	Binary Cross-Entropy
Patch size	256 × 256 pixels
Scheduler	None
Data augmentation	Rotation, flip, brightness jitter
Normalization	Per-band standardization (see Table 3)
Training framework	PyTorch Lightning

3.3.3 Experimental Setup and Computational Efficiency

The fine-tuning experiments were conducted using NVIDIA GeForce RTX 1080 Ti (for Prithvi) and NVIDIA GeForce RTX 1650 Super (for Satlas and MoE) GPUs. The training process for the MoE system, utilizing frozen experts, proved computationally efficient, requiring approximately 50 epochs to converge. Inference time for the complete MoE pipeline averages approximately 0.4 seconds per 256 × 256 patch on the RTX 1650 Super, demonstrating the operational feasibility of the approach for regional-scale mapping.

The full fine-tuning procedure for the Prithvi-EO-2.0 and Satlas models is summarized in Algorithm 1. It outlines the complete workflow from data loading to validation.

Algorithm 1. Fine-Tuning Procedure for Prithvi and Satlas Models

Input: Sentinel-2 patches $\mathbf{x}$, masks $\mathbf{y}$, pretrained model f_θ
Output: Fine-tuned expert model f_θ Split dataset into training/validation sets;
Split dataset into training/validation sets;
For each epoch **do**
 for each batch $(\mathbf{x}, \mathbf{y})$ **do**
 Resize $\mathbf{x}$ and $\mathbf{y}$ to 256×256 ;
 Apply augmentations;
 Prithvi: RandAugment-like (flip, rotate, brightness, contrast, gamma, noise);
 Satlas: Radiometric augmentation;
 Forward pass: $\hat{\mathbf{y}} = f_\theta(\mathbf{x})$;
 Compute loss: $\mathcal{L} = CE(\hat{\mathbf{y}}, \mathbf{y})$;
 Update θ using AdamW;
 Compute validation metrics: OA, IoU, F1, Precision, Recall;
 Apply EarlyStopping and save best checkpoint;
return Best parameters θ ;

3.4 Mixture of Experts

To potentially enhance performance beyond individual models, a MoE system was developed. MoE architectures are a powerful ensemble paradigm, originally introduced for efficiently scaling large neural networks by combining the strengths of multiple specialized sub-models (or “experts”).²⁴ The motivation for adopting an MoE approach in this study is primarily technical:

- **Complementary representations:** Prithvi (ViT-based) and Satlas (Swin Transformer-based) are built on distinct architectural principles, meaning they may capture different aspects of Sentinel-2 imagery. Leveraging both increases the diversity of learned features.
- **Robustness through ensembling:** Ensembles are known to reduce prediction variance and mitigate overfitting by combining multiple hypotheses, which is particularly valuable in limited-data regimes.
- **Dynamic weighting:** Unlike static averaging, the MoE framework employs a gating network that learns to adaptively assign weights to expert predictions depending on the input, enabling more flexible and context-aware decision-making.
- **Scalability:** MoE systems are computationally efficient in practice, as only a subset of experts is typically activated per input, making them attractive for scaling to larger models in future extensions.

The ensemble consists of the best-performing Prithvi and Satlas models, kept frozen as “experts”, and a lightweight, trainable CNN-based gating network.³⁷ The experts were kept frozen to preserve the robust feature representations learned during the individual fine-tuning stage and to avoid catastrophic forgetting given the limited dataset size.

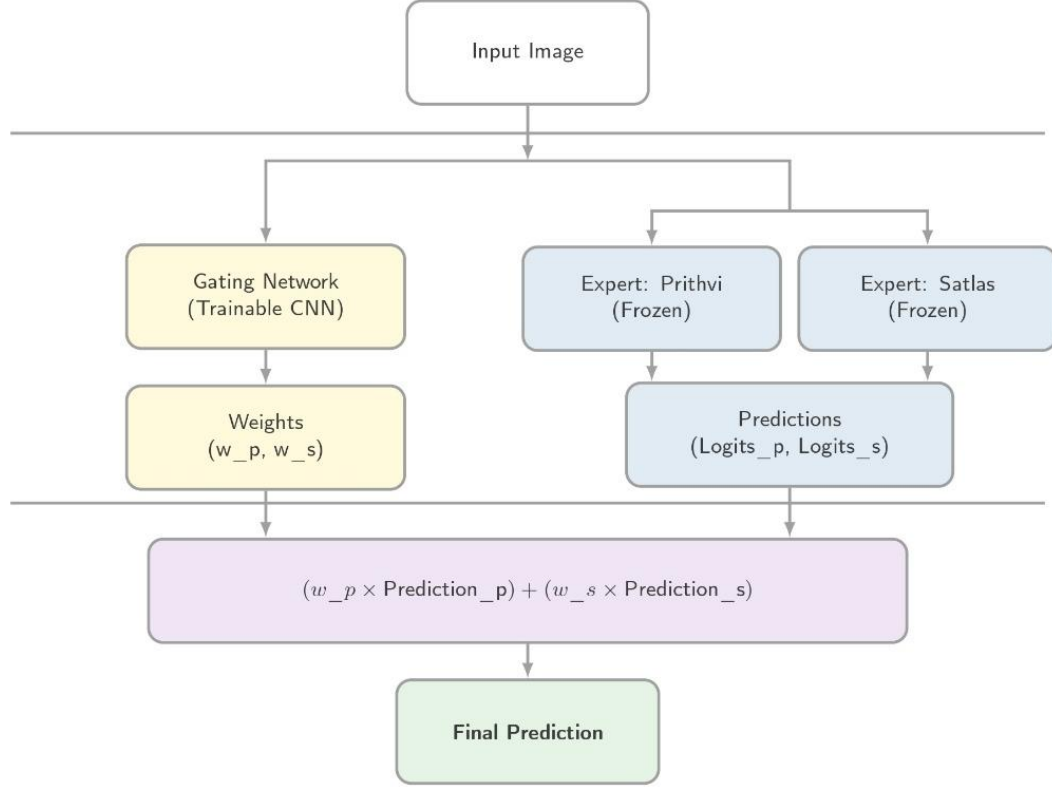


Fig. 6 Overall architecture of the proposed Mixture-of-Experts (MoE) system. The gating network receives the input Sentinel-2 patch and produces expert weights, which are used to compute a weighted combination of the Prithvi and Satlas expert predictions.

Formally, the final MoE prediction $\hat{y}$ for an input image $\mathbf{x}$ is obtained as a weighted combination of expert outputs (Eq. 4):

$$\hat{y} = \sum_{m=1}^M g_m(\mathbf{x}) f_m(\mathbf{x}), \quad (4)$$

Where $f_m(\mathbf{x})$ denotes the output of the m -th expert, $g_m(\mathbf{x})$ is the gating network weight for that expert, and $M = 2$ in this study (Prithvi and Satlas). The gating weights satisfy $\sum_{m=1}^M g_m(\mathbf{x}) = 1$, ensuring a convex combination of expert predictions.

The gating network dynamically assigns weights to the experts depending on the input image, enabling context-aware ensembling that outperforms naive averaging. To compute adaptive fusion weights for each expert, we employ a compact two-layer CNN followed by global pooling and a fully connected layer. The complete architecture of this gating network is summarized in Table 5.

The gating network was trained end-to-end while keeping the expert models strictly frozen. We utilized the AdamW optimizer with a learning rate of 1×10^{-4} and weight decay of 0.01. Training proceeded for a maximum of 50 epochs with a batch size of 1 (due to memory constraints of holding two experts) and Early Stopping monitored on validation loss with a patience of 10 epochs. To further improve robustness and generalization, the MoE system was trained using RandAugment, a recently proposed data augmentation technique. Unlike fixed augmentation pipelines, RandAugment randomly selects augmentation operations (e.g., rotation, translation, color jitter, contrast adjustment) from a predefined pool and applies them with varying intensity.³⁸ This stochasticity exposes the model to a wider diversity of transformations, reducing overfitting risks and enhancing the ability to generalize across unseen Sentinel-2 conditions and seasonal variations.

Table 5 Architecture of the gating network used to compute expert weights in the Mixture-of-Experts system.

Layer	Configuration	Output Shape
Input	6 input channels (Sentinel-2 bands)	$B \times 6 \times H \times W$
Conv2D + ReLU	16 filters, 3×3 , stride 1, padding 1	$B \times 16 \times H \times W$
MaxPool2D	2×2 kernel, stride 2	$B \times 16 \times H/2 \times W/2$
Conv2D + ReLU	32 filters, 3×3 , stride 1, padding 1	$B \times 32 \times H/2 \times W/2$
MaxPool2D	2×2 kernel, stride 2	$B \times 32 \times H/4 \times W/4$
Adaptive AvgPool2D	Output size: 1×1	$B \times 32 \times 1 \times 1$
Flatten	—	$B \times 32$
Fully Connected	Linear ($32 \rightarrow N_{\text{experts}}$)	$B \times N_{\text{experts}}$
Softmax	Per-expert normalized weights	$B \times N_{\text{experts}}$

Algorithm 2 describes how the frozen Prithvi and Satlas experts are combined through the gating network during inference.

Algorithm 2. Mixture-of-Experts (MoE) Inference and Gating

Input: 6-channels Sentinel-2 patches $\mathbf{x}$

Input: Frozen experts f_1 (Prithvi), f_2 (Satlas)

Output: Final fused logits $\hat{\mathbf{y}}$

Compute gating weights: $g = \text{softmax}(\text{GatingCNN}(\mathbf{x}))$;

Run Prithvi expert: $\mathbf{z}_1 = f_1(\mathbf{x})$;

Map $\mathbf{x}$ to Satlas 9-channel format: $\mathbf{x}_{\text{sat}} = T(\mathbf{x})$;

Run Satlas: $\mathbf{z}_2 = f_2(\mathbf{x}_{\text{sat}})$;

Fuse outputs:

$$\hat{\mathbf{y}} = g_1 \cdot \mathbf{z}_1 + g_2 \cdot \mathbf{z}_2$$

return $\hat{\mathbf{y}}$;

4 Results

Model performance was evaluated on the held-out validation set using a comprehensive suite of metrics : Intersection over Union (IoU), Overall Accuracy (OA), Precision, Recall, and the F1-score, following standard definitions commonly used in semantic segmentation.³⁹ The IoU quantifies the spatial overlap between predicted and reference cereal pixels, while OA captures the proportion of correctly classified pixels overall. Precision measures the reliability of cereal predictions (minimizing false positives), Recall measures the completeness of detections (minimizing false negatives), and the F1-score provides the harmonic mean of these two to assess balanced performance. All reported values are expressed as mean $\pm$ standard deviation across three independent runs.

4.1 Individual Model Performance

The Prithvi model was evaluated under four configurations, varying the data augmentation strategy in each case (baseline, geometric augmentations, radiometric augmentations, and RandAugment). The RandAugment configuration achieved the best overall results, reaching $81.42 \pm 0.48\%$ Overall Accuracy. However, the baseline Prithvi model achieved the highest Cereal-Class IoU (70.49%), indicating that the choice of augmentation affects general accuracy and class-specific overlap differently.

The Satlas model family demonstrated notably stronger performance. The best-performing variant was trained with subtle Radiometric Augmentation, achieving $96.93 \pm 0.31\%$ Overall Accuracy and a Cereal-Class IoU of $94.12 \pm 0.33\%$. This significantly surpassed the champion Prithvi configuration, as summarized in Table 6. Furthermore, Satlas achieved a high F1-score of $96.92 \pm 0.28\%$, reflecting highly consistent and reliable cereal discrimination.

Table 6 Comparative Performance : Champion Prithvi (RandAugment) vs. Champion Satlas (Radiometric Augmentation).

Metric	Prithvi (RandAugment)	Satlas (Radiometric Aug.)
Overall Accuracy (%)	81.42 ± 0.48	96.93 ± 0.31

IoU (Overall/Macro) (%)	68.65 ± 0.52	94.04 ± 0.36
IoU (Cereal Class) (%)	68.17 ± 0.47	94.12 ± 0.33
Precision (%)	81.43 ± 0.41	—
Recall (%)	81.49 ± 0.38	—
F1 Score (%)	81.41 ± 0.44	96.92 ± 0.28
Validation Loss	0.4146 ± 0.35	0.3427 ± 0.29

To better illustrate the training dynamics of the two champion models, Fig. 7 and Fig. 8 show the validation loss and accuracy trajectories across training epochs. Both models exhibit smooth and stable convergence, indicating that the optimization process was well-behaved. The Satlas model reaches a lower validation loss and maintains consistently higher validation accuracy throughout training, which aligns with the superior final IoU and F1 scores reported in Table 6. These curves further confirm that the performance gap between Prithvi and Satlas is not due to instability or underfitting, but rather stems from differences in their underlying representational capacity and pretraining strategies.

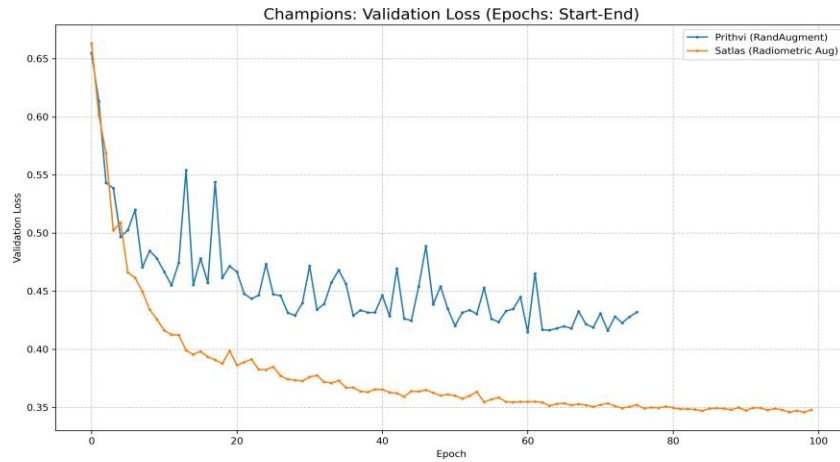


Fig. 7 Validation loss curves for the champion Prithvi (RandAugment) and Satlas (Radiometric Augmentation) models over training epochs. Both models exhibit stable convergence, with Satlas reaching consistently lower validation loss levels.

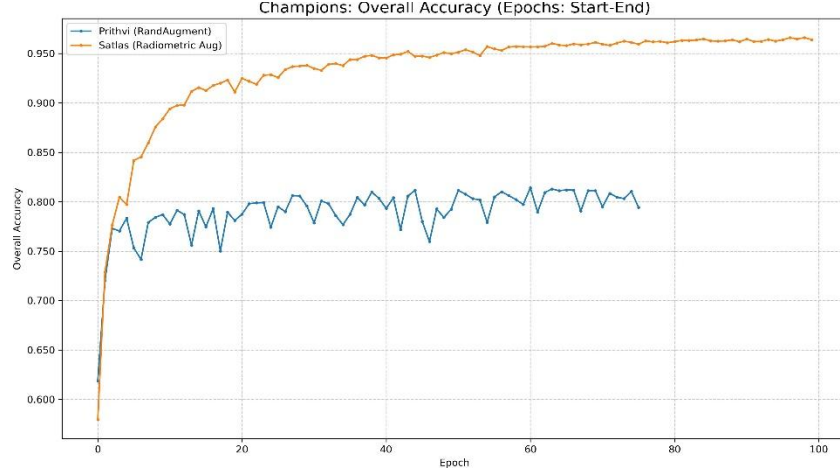


Fig. 8 Validation accuracy curves for the champion Prithvi (RandAugment) and Satlas (Radiometric Augmentation) models. Satlas achieves higher validation accuracy throughout training, reflecting its stronger feature representation and more effective pretraining.

4.2 Mixture of Experts (MoE) System Performance

The MoE system successfully leveraged the complementary strengths of the two champion models (Prithvi–RandAugment and Satlas–Radiometric Augmentation), achieving the highest performance across all reported metrics. As detailed in Table 7, the MoE obtained an Overall Accuracy of $97.82 \pm 0.26\%$ and a Cereal-Class IoU of $95.58 \pm 0.29\%$, representing a consistent improvement over the standalone Satlas champion, particularly in reducing prediction variance.

The MoE further achieved a Precision of $96.39 \pm 0.30\%$, Recall of $96.44 \pm 0.27\%$, and an F1-score of $96.41 \pm 0.28\%$, indicating a highly consistent balance between detection reliability and completeness.

Analysis of the gating network revealed that it rapidly converged toward assigning higher weights to the Satlas expert—reflecting its stronger standalone performance—while still utilizing the Prithvi expert to refine difficult or ambiguous regions. This demonstrates that the MoE fusion mechanism effectively exploits cross-model complementarities rather than merely replicating the best single expert.

Table 7 Comparative Performance : MoE System vs. Champion Models.

Metric	Prithvi (RandAugment)	Satlas (Radiometric Aug.)	MoE System
Overall Accuracy (%)	81.42 ± 0.48	96.93 ± 0.31	97.82 ± 0.26
IoU (Cereal Class) (%)	68.65 ± 0.52	94.04 ± 0.36	95.58 ± 0.29

Precision (%)	81.43 ± 0.41	—	96.39 ± 0.30
Recall (%)	81.49 ± 0.38	—	96.44 ± 0.27
F1 Score (%)	81.41 ± 0.44	96.92 ± 0.28	96.41 ± 0.28
Validation Loss	0.4146 ± 0.35	0.3427 ± 0.29	0.3341 ± 0.31

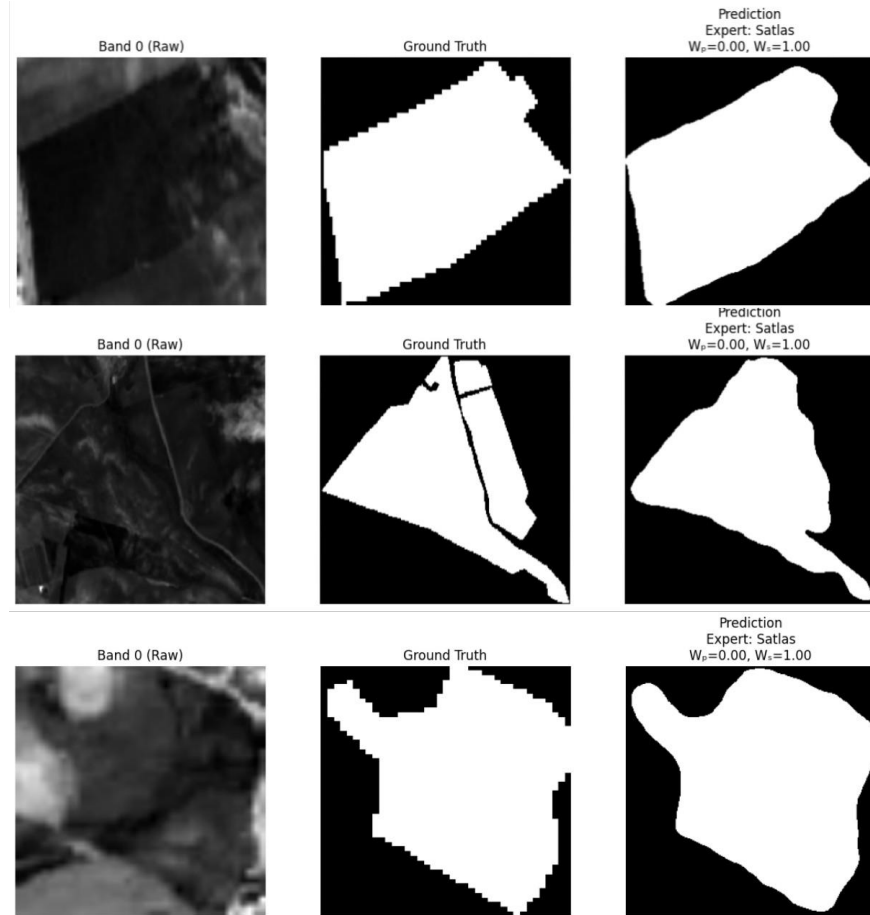


Fig. 9 Qualitative segmentation result from the final MoE model on a sample validation patch. From left to right: the input image, the ground truth mask, and the model’s predicted segmentation mask.

To contextualize these findings, we next compare our results with recent state-of-the-art studies.

5 Discussion

The comparative performance of Prithvi and Satlas can be understood through documented differences in pretraining strategy, data scale, and architectural design. In this study, we prioritized comparing these Foundation Models against each other to assess the specific value of pre-training in data-scarce domains. Traditional deep learning models (e.g., U-Net or ResNet trained from scratch) were not included as baselines because prior literature [9,10](#) confirms they fail to generalize

on extremely small datasets (<150 samples) without extensive pre-training. Consequently, the single-expert foundation models serve as the most rigorous state-of-the-art baselines for this study. Prithvi-EO-2.0 is explicitly described as a ViT-based model pretrained using a masked autoencoder (MAE) self-supervised objective over large multi-temporal Sentinel-2 and Landsat image collections.¹⁶ This pretraining strategy focuses on reconstructive representation learning from unlabeled data, producing general purpose spectral–spatial features.

SatlasPretrain, in contrast, is grounded in a large-scale labeled corpus containing over 302 million geospatial labels across 137 categories.¹⁵ The Satlas project also provides pretrained models built using a backbone–FPN–task-head architecture designed for dense prediction tasks such as semantic segmentation.^{40,41} These characteristics indicate that Satlas benefits from semantically rich supervisory signals and architectural components explicitly suited for pixel-level discrimination.

These documented differences offer a clear explanation for Satlas outperforming Prithvi in the few-shot cereal-mapping task: Satlas leverages large-scale labeled pretraining and dense-prediction architectures, whereas Prithvi’s MAE-based self-supervision yields strong but less class-specialized features without additional extensive fine-tuning. This interpretation is consistent with both the published model descriptions and the observed empirical gap.

Within the Mixture-of-Experts system, the gating network consistently assigned higher weights to the Satlas expert across validation samples, while still drawing selectively on Prithvi predictions in more ambiguous regions. This architectural diversity—combining a ViT-based MAE with a Swin-based supervised model—acts as a regularization mechanism, preventing the ensemble from over-fitting to artifacts specific to a single backbone. This behavior—directly observed in the gating outputs—indicates that the MoE mechanism effectively exploits complementary representations rather than simply replicating the stronger expert. Overall, the performance patterns align with the known characteristics of the two model families and provide an evidence-based rationale for the superiority of the Satlas and MoE configurations in this setting.

5.1 Comparison with Recent Crop-Mapping Studies

Table 8 Comparison with selected recent crop-mapping studies.

Study	Region / Data	Method	Metric	Reported Value
Our study	Sétif, Algeria / (Sentinel-2)	MoE (Prithvi + Satlas)	IoU (cereal); OA	95.58%; 97.82%

Mohammadi et al., 2024 ⁴³	France / Ghana (S2 time series)	Few-shot learning (-TIM)	Macro-F1	75.9% (FR); 59.6% (GH)
Gao et al., 2023 ⁴²	USA (Sentinel-2)	A2SegNet	OA; mIoU gain	91.5–94.2% OA; +6.6% mIoU
Lu et al., 2023 ⁴³	Sentinel-2	MSSNet	OA; mIoU	90.77% OA; 76.59 mIoU
Yan et al., 2025 ⁴⁴	PASTIS-R benchmark	Spatiotemporal GFM	mIoU; F1	84–89% mIoU ; Up to 92% F1
Li et al., 2025 ⁴⁵	Multi-sensor (MODIS, L8/9, S2)	Multi-source temporal FM	mIoU	83–88% mIoU on downstream crop tasks
Zhu et al., 2024 ⁴⁶	SITS segmentation	Weakly-sup. temp. net	IoU-equiv.; F1	90–94% IoU-equiv.; 92% F1

As shown in Table 8, our MoE ensemble achieves a cereal IoU of 95.58% and an overall accuracy of 97.82%, placing it among the strongest recent results in the literature. To enable fairer comparison, metrics are harmonized where possible (IoU or mIoU for segmentation tasks, F1 or Macro-F1 for classification-based studies).

Recent foundation-model approaches such as CropSTS⁴⁴ and AgriFM⁴⁵ report mIoU values in the 83–89% range on multi-class cropland benchmarks, while weakly supervised SITS models such as Exact⁴⁶ reach approximately 90–94% IoU-equivalent accuracy. Traditional deep models such as A2SegNet⁴² and MSSNet⁴³ typically fall within 76–94% mIoU/OA, depending on region and crop type. Few-shot methods (e.g., Mohammadi et al.¹³) obtain 59–76% Macro-F1 under low-supervision scenarios.

Although direct numerical comparison is limited by differences in class definitions, geographic regions, and sensor inputs, these harmonized metrics indicate that our MoE system is highly competitive with both classical and recent foundation-model-based crop mapping approaches. The high cereal IoU obtained in this study highlights the benefit of expert fusion for fine-grained agricultural segmentation tasks.

6 Conclusion

This study evaluated two state-of-the-art Earth observation foundation models—Prithvi-EO-2.0 and SatlasPretrain—for few-shot cereal crop mapping using Sentinel-2 imagery, and introduced a lightweight Mixture-of-Experts (MoE) fusion strategy to combine their complementary strengths. Across three independent runs, the Satlas model consistently outperformed all Prithvi variants, achieving an Overall Accuracy of $96.93 \pm 0.31\%$ and a Cereal-Class IoU of $94.12 \pm 0.33\%$. The proposed MoE system further improved performance, reaching $97.82 \pm 0.26\%$ Overall Accuracy, a Cereal-Class IoU of $95.58 \pm 0.29\%$, and an F1-score of $96.41 \pm 0.28\%$. These results demonstrate that integrating expert models through a small gating CNN can yield measurable gains over individual foundation models, even under limited-label conditions.

The study also highlights an important insight for agricultural remote sensing: models pre-trained with large, label-rich geospatial corpora (e.g., SatlasPretrain) appear to transfer more effectively to fine-grained semantic tasks than models relying solely on self-supervised reconstruction objectives (e.g., Prithvi-EO-2.0). The MoE framework successfully leveraged these complementary representations, providing a robust pathway for improving performance without extensive retraining. Taken together, the findings indicate that foundation-model fusion represents a promising direction for reliable, scalable cereal crop mapping in data-scarce environments.

7 Limitations and Future Work

While the proposed system demonstrates strong performance within the scope of the present study, several limitations must be acknowledged. First, all experiments were conducted on a relatively small dataset collected entirely within a single agro-ecological region (Sétif, Algeria). Although the results indicate that both Satlas and the MoE system achieve high accuracy in this context, the restricted geographic extent limits the ability to assess model robustness under different climate

regimes, soil backgrounds, crop management practices, and illumination conditions. This constitutes a core limitation of the current evaluation. As part of future work, we plan to perform a comprehensive cross-region validation by incorporating additional sites with distinct environmental characteristics to more rigorously test model generalization.

Second, the study focuses exclusively on Sentinel-2 imagery. While this choice ensures spectral richness and operational accessibility, it does not guarantee transferability to other optical sensors or to platforms with different spatial or radiometric properties. Evaluating the system under alternative sensors such as Landsat-8/9, PlanetScope, or commercial very-high-resolution imagery represents an important future direction for assessing sensor-agnostic performance.

Third, the available dataset displays class imbalance, with background pixels greatly outnumbering cereal pixels. Although the use of IoU, Recall, and F1-score mitigates some of the bias introduced by imbalance, the training process may still favor majority classes. Future work may incorporate class-balanced loss functions, focal loss variants, or targeted sampling strategies to reduce this effect.

Fourth, although the MoE system demonstrated clear improvements, its scalability to larger numbers of experts or higher spatial resolutions has not been tested. Investigating whether the gating network remains effective with additional models or under operational-scale deployments is an open question that warrants further study.

In summary, the present work should be viewed as a promising region-specific investigation. Future efforts will prioritize (i) cross-region validation across multiple agricultural zones, (ii) evaluation under diverse sensors, (iii) improved handling of class imbalance, and (iv) examining the scalability and operational deployment of the Mixture-of-Experts framework.

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Disclosures

The authors declare that there are no financial interests, commercial affiliations, or other potential conflicts of interest that could have influenced the objectivity of this research or the writing of this paper.

Code, Data, and Materials Availability

The source code, models, and dataset generated and analyzed during the current study are openly available in a GitHub repository at the following URL: <https://github.com/Amir-001/Final-Year-Project>. and a Zenodo DOI: <https://doi.org/10.5281/zenodo.17674368>

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Caption List

Fig. 1 General workflow of the proposed methodology.

Fig. 2 The primary study area in Sétif, Algeria. The ground truth cereal parcel polygons collected for this research are highlighted in red.

Fig. 3 Examples of a training samples, showing the input Sentinel-2 image patch (left) and its corresponding binary ground truth mask (right), where white represents the 'cereal' class.

Fig. 4 Architecture of Prithvi-EO-2.0. The model uses ViT-based masked autoencoder pretraining with 3D patch embeddings to jointly encode spatial and temporal structure. Geolocation and temporal metadata are added through learned positional encodings.

Fig. 5 Architecture of the SatlasPretrain model. The backbone extracts spatial features from each input image, an FPN builds multi-scale feature representations, and the prediction head produces dense outputs. Multi-image inputs are aggregated using temporal max pooling.

Fig. 6 Overall architecture of the proposed Mixture-of-Experts (MoE) system. The gating network receives the input Sentinel-2 patch and produces expert weights, which are used to compute a weighted combination of the Prithvi and Satlas expert predictions.

Fig. 7 Validation loss curves for the champion Prithvi (RandAugment) and Satlas (Radiometric Augmentation) models over training epochs. Both models exhibit stable convergence, with Satlas reaching consistently lower validation loss levels.

Fig. 8 Validation accuracy curves for the champion Prithvi (RandAugment) and Satlas (Radiometric Augmentation) models. Satlas achieves higher validation accuracy throughout training, reflecting its stronger feature representation and more effective pretraining.

Fig. 9 Qualitative segmentation result from the final MoE model on a sample validation patch. From left to right: the input image, the ground truth mask, and the model's predicted segmentation mask.

Table 1 Recent works (2023–2025) related to foundation models, few-shot learning, and ensemble/MoE and diffusion methods in remote sensing.

Table 2 Study area metadata.

Table 3 Per-band reflectance statistics used for normalization.

649 **Table 4** Training hyperparameters used for fine-tuning the Prithvi and Satlas models.
650 **Table 5** Architecture of the gating network used to compute expert weights in the Mixture-of-
651 Experts system.
652 **Table 6** Comparative Performance : Champion Prithvi (RandAugment) vs. Champion Satlas
653 (Radiometric Augmentation).
654 **Table 7** Comparative Performance: MoE System vs. Champion Models.
655 **Table 8** Comparison with selected recent crop-mapping studies.